

STADIOalot Returns Prediction

Module: Capstone Project

Assignment: SS2 – Part D

Recommendations Report

Student Name: Brondon-Lee Paulus

Student Number: 26303892

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1. Which model is best between Model 1 and Model 2?

Based on the results presented in Part C, **Model 1, Logistic Regression, is the better model** for this project. Both models were tested using the same test dataset and the same performance metrics, which makes the comparison fair.

Logistic Regression achieved an accuracy of **88.05%**, while Random Forest achieved **85.31%**. Logistic Regression also achieved a higher precision of **89.19%**, F1 score of **93.32%** and ROC-AUC of **78.25%**. Random Forest achieved a precision of 85.31%, F1 score of 92.07% and ROC-AUC of 77.60%.

Random Forest did achieve a recall of 100%, which was higher than the 97.84% achieved by Logistic Regression. However, this result needs to be considered carefully. The confusion matrix showed that Random Forest predicted every observation as a Positive Outcome. This meant that it correctly identified all 16,930 Positive Outcomes, but it failed to identify any of the 2,915 Negative Outcomes.

Logistic Regression performed better in this area because it identified **908 Negative Outcomes**. Although it still incorrectly classified 2,007 Negative Outcomes as Positive Outcomes, it was able to identify observations from both classes. This makes it more useful than Random Forest for this dataset.

The results also show why accuracy should not be used on its own. The test dataset was imbalanced, with Positive Outcomes making up **85.31%** of the observations. A model that mainly predicts the majority class can therefore achieve a high accuracy without actually being useful for identifying the minority class.

Based on the accuracy, precision, F1 score, ROC-AUC and confusion matrix, **Logistic Regression provides the stronger baseline model** and should be taken forward for further development.

2. How can the selected model be improved?

The main area for improvement is the model's ability to identify the minority Negative Outcome class. Logistic Regression correctly identified 908 Negative Outcomes, but it incorrectly classified 2,007 Negative Outcomes as Positive Outcomes. This shows that the model can be improved further.

One possible improvement would be to address the class imbalance in the training data. Techniques such as class weighting or resampling could be tested to give the minority class more importance during training. This could help the model identify more Negative Outcomes instead of mainly learning the majority class.

The classification threshold could also be adjusted. The current model uses a standard classification threshold, but testing different thresholds could improve the balance between precision and recall. This would be useful if identifying potential negative outcomes is more important to the business than achieving the highest overall accuracy.

Feature engineering should also be used to improve the model. Mishra and Dutta (2024) found that product, order and transaction variables can be useful when predicting customer outcomes. Therefore, additional features such as product category, price, payment information, delivery information and customer purchasing behaviour could be tested.

Hyperparameter tuning and cross-validation should also be included in future development. Different Logistic Regression settings could be tested to determine whether the model's F1 score and ROC-AUC can be improved. The confusion matrix should continue to be reviewed because it provides a clearer indication of how the model performs on both classes.

3. How will the selected model need to be adapted to work with the requested SS1 data?

The current model is a proof-of-concept using the public Olist dataset. The Olist dataset does not contain a direct product-return variable, so the project used customer review scores as a proxy outcome. The model would therefore need to be adapted before it could be used with the actual STADIOalot data requested in SS1.

The most important change would be to replace the proxy outcome with an **actual product-return outcome**. The model should be trained using historical STADIOalot information showing whether a product was returned or not. This would make the target variable directly related to the business problem.

The additional data requested in SS1 could also provide more useful features. These could include customer information, product information, order information, transaction details and delivery information. Product category, price, delivery time and customer purchasing behaviour could potentially help the model identify patterns associated with returns.

The preprocessing process would also need to be adapted to the STADIOalot data. Categorical variables would need to be converted into a format that Logistic Regression can use, numerical variables may need to be scaled and missing values would need to be handled appropriately.

It would also be important to prevent **data leakage**. Only information that would be available before the return occurs should be used as a prediction feature. For example, information created after a product has already been returned should not be included when training the model.

Once the STADIOalot data is available, Logistic Regression could be retrained using the actual historical return data. The model could then be tested on unseen data to determine whether the performance achieved using the public dataset can also be achieved in the real business environment.

4. Do the results align with the related literature from Part A?

The results from Part C are generally aligned with the related literature from Part A, although there are some important differences between the studies and this project.

Heilig et al. (2016) used the DMC 2014 dataset to investigate product returns and compared different machine learning approaches, including Logistic Regression, Random Forest and other models. Their study showed that machine learning can be used to predict product returns and that comparing different models is important. The results from Part C also support this because two different models were tested and Logistic Regression performed better overall than Random Forest.

The results also have similarities with Mishra and Dutta (2024), who compared Logistic Regression, Random Forest and Gradient Boosting for product-return prediction. Their study used ROC-AUC as one of the measures for comparing the models. In this project, Logistic Regression achieved a slightly higher ROC-AUC of **78.25%**, compared with **77.60%** for Random Forest. This supports the use of ROC-AUC when comparing classification models.

Mishra and Dutta (2024) also highlighted the importance of product, order and transaction variables. This supports the recommendation to include more of these variables when the STADIOalot data becomes available.

The findings are also consistent with Duong et al. (2025), who found that product characteristics can provide useful information when predicting product return behaviour. This supports the use of product-related information as part of the future feature engineering process.

However, the results should not be considered exactly the same as the previous studies. The current Part C experiment used the **Olist public dataset**, where customer review scores were used as a proxy outcome. The studies in Part A used different datasets and, in some cases, actual product-return outcomes. Therefore, the results from Part C demonstrate that the approach is viable as a proof-of-concept, but the model still needs to be tested using the actual STADIOalot return data.

Conclusion

Based on the Part C results, **Logistic Regression is recommended as the baseline model**. It achieved better accuracy, precision, F1 score and ROC-AUC than Random Forest and, importantly, it was able to identify some of the minority Negative Outcomes.

The model can be improved by addressing class imbalance, adjusting the classification threshold, performing further feature engineering and tuning the model. When the requested STADIOalot data becomes available, the proxy outcome should be replaced with an actual product-return outcome and the model should be retrained using relevant customer, product, order and delivery information.

Overall, the results are broadly consistent with the related literature from Part A. However, the difference between the Olist dataset and the datasets used in the literature means that further testing with actual STADIOalot data will be required before the model can be used in a real business environment.

References

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